# Pratik Nanaso Raut

Pune, Maharashtra | +91 8600475940 | pratikraut.codex@gmail.com | https://linkedin.com/in/pratik-raut-85248b319 | https://github.com/pratik-codeX

## PROFESSIONAL SUMMARY

Computer Engineering candidate with strong foundation in Java, Spring Boot, REST APIs — backend development. Hands-on experience developing ERP modules using Java, Spring Boot, Hibernate, and REST APIs through practical projects.

## TECHNICAL SKILLS

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| **Languages** | Java, C, C++, SQL, JavaScript, Python (basics) |
| **Backend** | Spring Boot, REST APIs, Spring Security, Hibernate, JWT, JDBC |
| **Databases** | MySQL, MongoDB |
| **Tools** | Git, GitHub, Maven, Postman, IntelliJ IDEA |
| **Concepts** | SDLC, Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), REST API Design, Design Patterns, Problem Solving, Database Optimization, API Security, Microservice Fundamentals, Time Complexity Analysis, RBAC, Multi-Tenant Architecture, JWT Authentication, Multithreading |

## PROFESSIONAL EXPERIENCE

**Software Developer Intern** *Feb 2026 -- Ongoing*
*SetTribe IT Solutions | Pune*

- Developed and enhanced ERP backend modules using Java, Spring Boot, Hibernate, and REST APIs for employee and operational workflows.
- Integrated PostgreSQL database operations, validation logic, and API testing workflows using Postman.
- Collaborated in Agile sprint cycles, performing Git-based code reviews, bug fixes, feature implementation, and deployment support.

## ACADEMIC PROJECTS

**Customised Virtual File System (CVFS)** -- In-Memory Linux-like File Manager *C, C++, Data Structures, OS Concepts*

- Simulated Linux file system operations using C++ and inode-based metadata management.
- Achieved O(1) file lookup by designing hash table-based inode indexing.
- Implemented in-memory file allocation table managing file permissions and directory traversal.

**ProcMemInspector** -- Linux Process Memory Layout Analyzer *C, Linux Internals, /proc Filesystem*

- Developed /proc filesystem parser in C to display virtual memory layout of processes.
- Implemented process statistics monitoring by reading /proc/[pid]/maps and /proc/[pid]/status.
- Analyzed virtual address space segmentation demonstrating memory isolation.

**Concurrent FTP Server** -- Multi-Client File Transfer Protocol Server *C, POSIX Threads, TCP Sockets, Multithreading*

- Engineered concurrent client handling using POSIX threads over TCP sockets.
- Implemented thread-safe resource management using mutex and semaphore primitives.
- Designed application-layer protocol over raw sockets handling authentication and commands.

## TECHNICAL STRENGTHS

- Strong foundation in C++ OOP, STL containers, templates, and dynamic memory management.
- Proficient in POSIX system calls, concurrent programming, and Linux kernel interface design.
- Hands-on experience implementing OS-level components: file systems, network servers, and memory analyzers.
- Practical exposure to network programming, socket communication, and concurrent server development.

## EDUCATION

**Master of Computer Applications (MCA)** *2026 | Pursuing*
Dnyansagar Institute of Management and Research, Pune (SPPU) 7.24 CGPA **Bachelor of Computer Applications (BCA)** *2024*
C.T. Bora College, Shirur (SPPU) 73.34% **Higher Secondary Certificate (HSC)** *2021*
Maharashtra State Board 75%